Improve Generalization

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One current controlled-improvement record.

Trace one data-version change,compare all outputs and design source-aware next evidence.

Keep one habit: Fresh W1–W3/W5fields → selected blue/white plan → project repair record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Overfitting**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Feature Engineering

> **My remembered idea, example and correction**

Overfitting

> **My remembered idea, example and correction**

Train, Validation, Test

> **My remembered idea, example and correction**

W1 · Declare the controlled experiment

> Fresh controlled-improvement responses preserve the earlier capacity-practice fields as review.

Complete current trial reference

Current trial V1T1–T4 hasbackground/markererrors 0/0, background-first tie choosesbackground. V2 addsT5/T6 marker1on grey andT7/T8 marker0on red; errors 4/0 choosemarker. Fixed validationP1 spotted/P2 healthy, eachred/grey:V1 gets2/4 views, V2 gets4/4; only 2 newplants. Same learner, rule, candidate order/tie, validation references and membership; training version changed. Perfect marker alignment is synthetic.

Separate shift fixturebright 2/20=10%, dim 6/20=30%, pooled 8/40=20%; neither pooled result nor condition association proves cause.

Current task and response

- Name the changed factor and at least four fixed factors in the complete T/Vtrial. Explain the training-only candidate/tie rule.

> **My controlled-improvement reasoning and evidence**

▼ **Review · Earlier capacity task and saved response**

Reference for this exercise: open this week's student lesson, "Improve Generalization", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace before and after

> Fresh controlled-improvement responses preserve the earlier capacity-practice fields as review.

Complete current trial reference

Current trial V1T1–T4 hasbackground/markererrors 0/0, background-first tie choosesbackground. V2 addsT5/T6 marker1on grey andT7/T8 marker0on red; errors 4/0 choosemarker. Fixed validationP1 spotted/P2 healthy, eachred/grey:V1 gets2/4 views, V2 gets4/4; only 2 newplants. Same learner, rule, candidate order/tie, validation references and membership; training version changed. Perfect marker alignment is synthetic.

Separate shift fixturebright 2/20=10%, dim 6/20=30%, pooled 8/40=20%; neither pooled result nor condition association proves cause.

Current task and response

- For the complete T1–T8 and P1/P2 fixture, explain V1 and V2 candidate training errors, chosen feature and every validation prediction. Report both view and plant counts. State what was held fixed, what changed and why 4/4 development views do not prove general reliability.

My controlled-improvement reasoning and evidence

▼ **Review · Earlier capacity task and saved response**

Does a **training**-validation gap alone prove **overfitting**?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair a perfect-development claim

> Fresh controlled-improvement responses preserve the earlier capacity-practice fields as review.

Complete current trial reference

Current trial V1T1–T4 hasbackground/markererrors 0/0, background-first tie choosesbackground. V2 addsT5/T6 marker1on grey andT7/T8 marker0on red; errors 4/0 choosemarker. Fixed validationP1 spotted/P2 healthy, eachred/grey:V1 gets2/4 views, V2 gets4/4; only 2 newplants. Same learner, rule, candidate order/tie, validation references and membership; training version changed. Perfect marker alignment is synthetic.

Separate shift fixturebright 2/20=10%, dim 6/20=30%, pooled 8/40=20%; neither pooled result nor condition association proves cause.

Current task and response

- A team adds crossed training data and inspects 4/4 correct validation views from 2 plants. Does that establish robust performance on new species, cameras or backgrounds? Name the missing evidence.

My controlled-improvement reasoning and evidence

▼ **Review · Earlier capacity task and saved response**

Describe one fair next development comparison.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the fresh evaluation

> Fresh controlled-improvement responses preserve the earlier capacity-practice fields as review.

Complete current trial reference

Current trial V1T1–T4 hasbackground/markererrors 0/0, background-first tie choosesbackground. V2 addsT5/T6 marker1on grey andT7/T8 marker0on red; errors 4/0 choosemarker. Fixed validationP1 spotted/P2 healthy, eachred/grey:V1 gets2/4 views, V2 gets4/4; only 2 newplants. Same learner, rule, candidate order/tie, validation references and membership; training version changed. Perfect marker alignment is synthetic.

Separate shift fixturebright 2/20=10%, dim 6/20=30%, pooled 8/40=20%; neither pooled result nor condition association proves cause.

Current task and response

- Design a next collection/evaluation that tests the frozen revision on new source plants and declared intended-use combinations. State what results are still unknown and why the inspected P1/P2 views cannot be called untouched final evidence.

My controlled-improvement reasoning and evidence

▼ **Review · Earlier capacity task and saved response**

What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected design task uses blue/white training backgrounds, not the completed red/grey toy trial. The training correlation alone does not prove reliance on background. Plan same-leaf crossed-background diagnostics on development sources, one justified training revision, and a separate fresh grouped final evaluation after choices are frozen.

  No observed blue/white predictions, benefit or final metrics are supplied; record them as proposed rather than invent results.

- Spotted leaves have blue backgrounds and healthy leaves white backgrounds in **training**. Design a generalisation investigation with fresh evaluation.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Improve Plant Classifier with one approved change and a full before/after **evidence** table.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Earlier **capacity**-practice completion record:retain the saved response as review. Use current W2for the controlled original/crossed training-data trial.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A classifier relies on tray colour; new trays reduce performance.
- Propose controlled improvement.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A team proposes more representative **training** examples and a simpler model. How should it compare changes while protecting the final test?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 30 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

▼ **Extension · Distribution shift — W1 Compare conditions**

REFERENCE · SECTION 17

Extension · Distribution shift — W1 Compare conditions

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Calculate each worked error rate and the difference.

> **My reasoning and evidence**

▼ **Extension · Distribution shift — W2 Read pooled evidence**

REFERENCE · SECTION 18

Extension · Distribution shift — W2 Read pooled evidence

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Why is 8/40 incomplete by itself?

My reasoning and evidence

▼ Extension · Distribution shift — W3 Wait for labels

REFERENCE · SECTION 19

Extension · Distribution shift — W3 Wait for labels

Retained extension task with its own values and teacher key.

Extension question and evidence

- Can changed **prediction**s alone prove an accuracy drop?

My reasoning and evidence

▼ Extension · Distribution shift — W4 Apply independently

REFERENCE · SECTION 20

Extension · Distribution shift — W4 Apply independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Old-like data has 3 errors/30 and new-condition data 5/10. Give both rates and the pooled rate.

> **My reasoning and evidence**

▼ **Extension · Distribution shift — W5 Plan a response**

REFERENCE · SECTION 21

Extension · Distribution shift — W5 Plan a response

> Retained extension task with its own values and teacher key.

Extension question and evidence

- State a safe classroom response to **input**s outside the supported range.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Test whether a plant model follows the leaf or a convenient background shortcut.
- Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Test whether a plant model follows the leaf or a convenient background shortcut.
- Which labelled step matches this detail? Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.
- 1. Notice correlation
- 2. Design crossed cases

- 3. Inspect fixed model

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Inspect fixed model Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. Read the picture in order.

Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Which conclusion is supported by the supplied facts?

- 1. It fails on spotted/grey and healthy/red in the crossed audit.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Repair and keep the record

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Overfitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model **capacity**. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

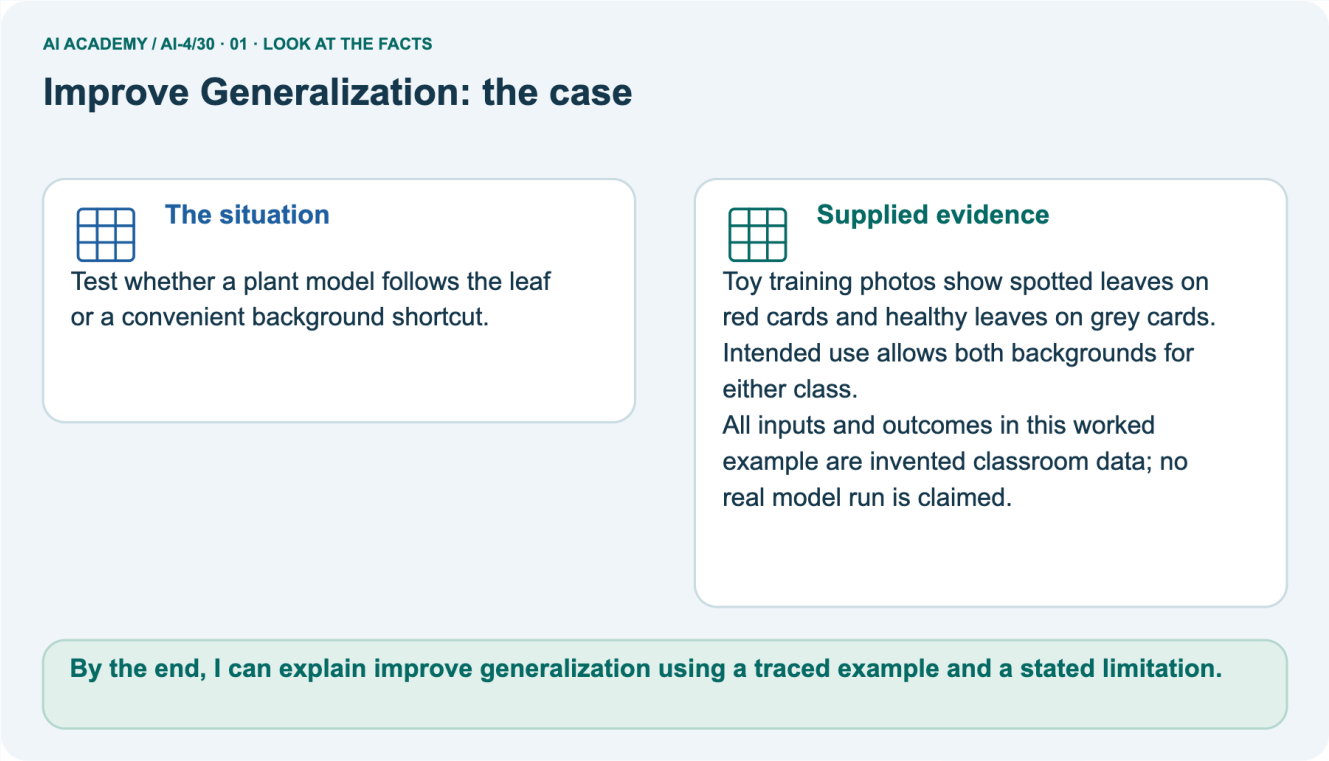
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Improve Generalization: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

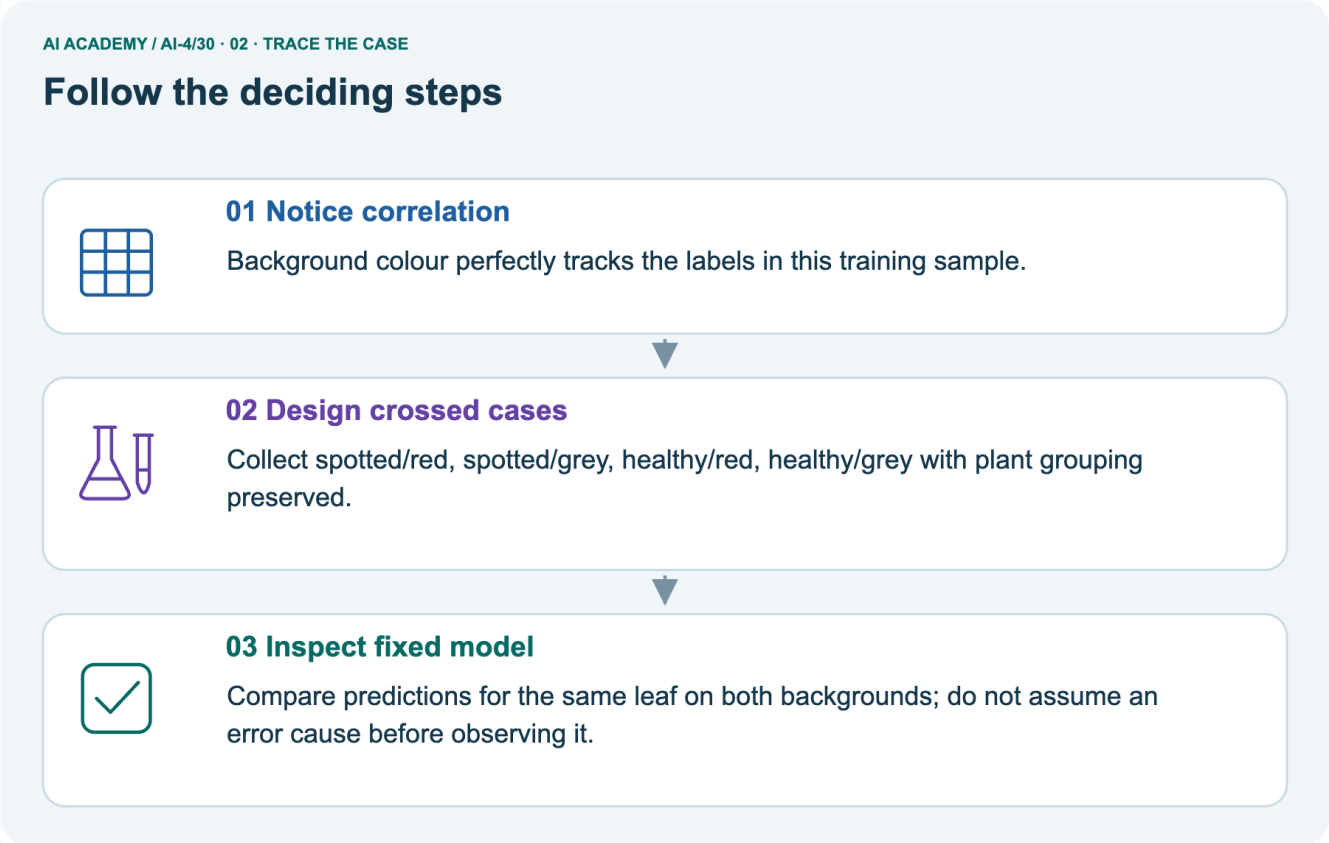
Test whether a plant model follows the leaf or a convenient background shortcut. Supplied facts: Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards.

Intended use allows both backgrounds for either class. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Notice correlation: Background colour perfectly tracks the labels in this **training** sample. → Design crossed cases: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved. → Inspect fixed model: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Overfitting and model capacity at a glance

AI ACADEMY / AI-4/30 · 03 · INSPECT THE LESSON MODEL

Overfitting and model capacity at a glance

Part	Evidence or action	Interpretation
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

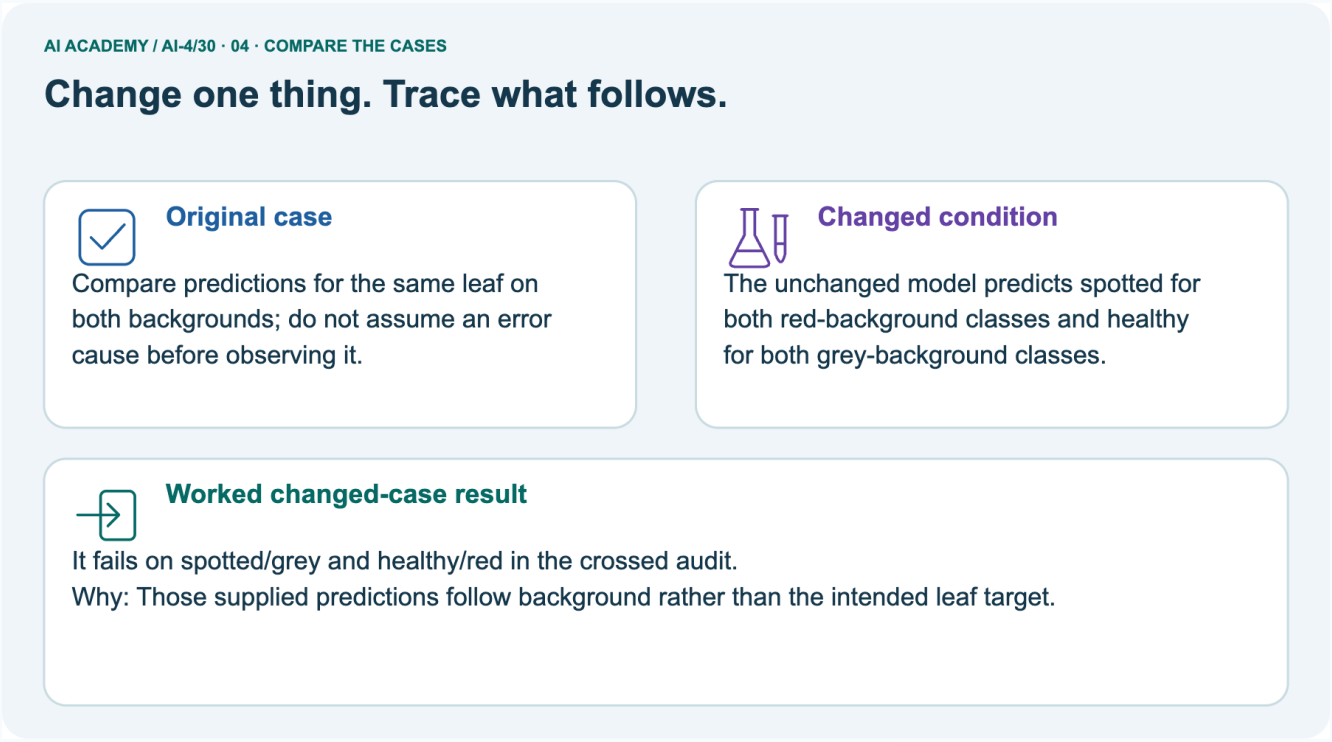
Follow the labelled picture with the matching explanation. Open the vector diagram

Overfitting and model **capacity** at a glance. Constant | **Training** 5 | Validation 6. Line | **Training** 2 | Validation 3. Flexible curve | **Training** 0 | Validation 8

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

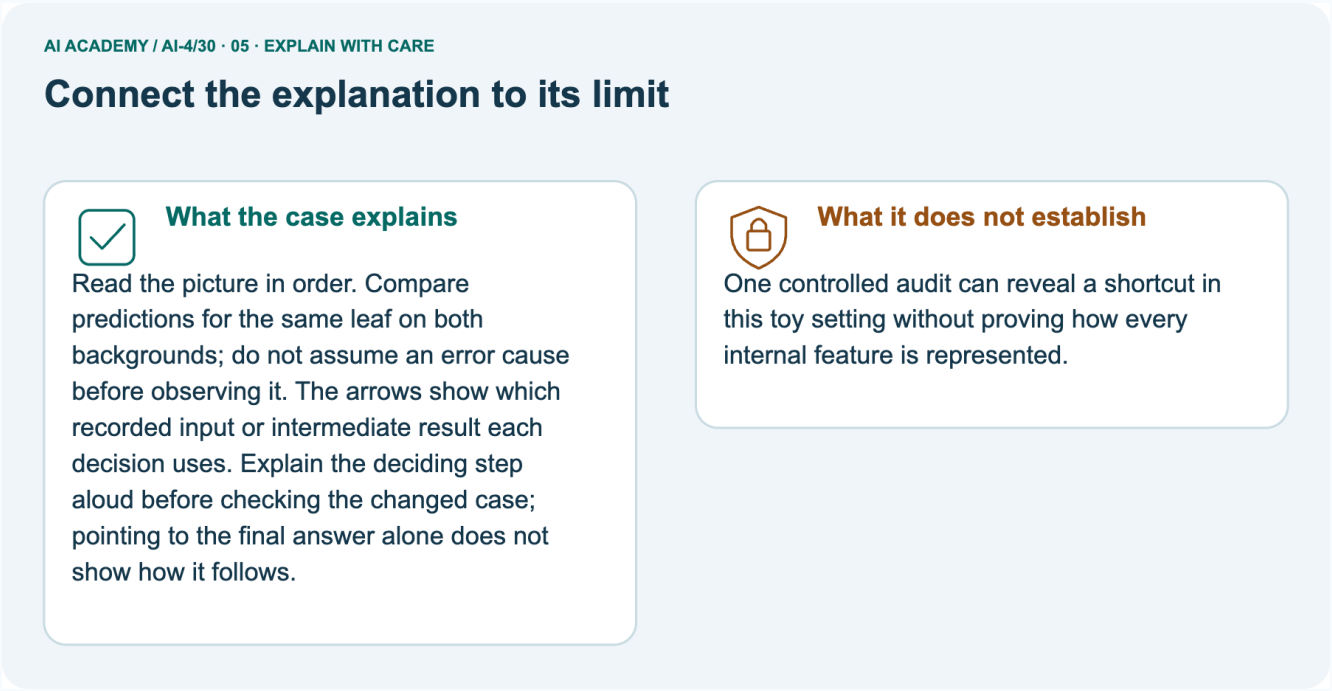
Changed condition: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Expected result: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

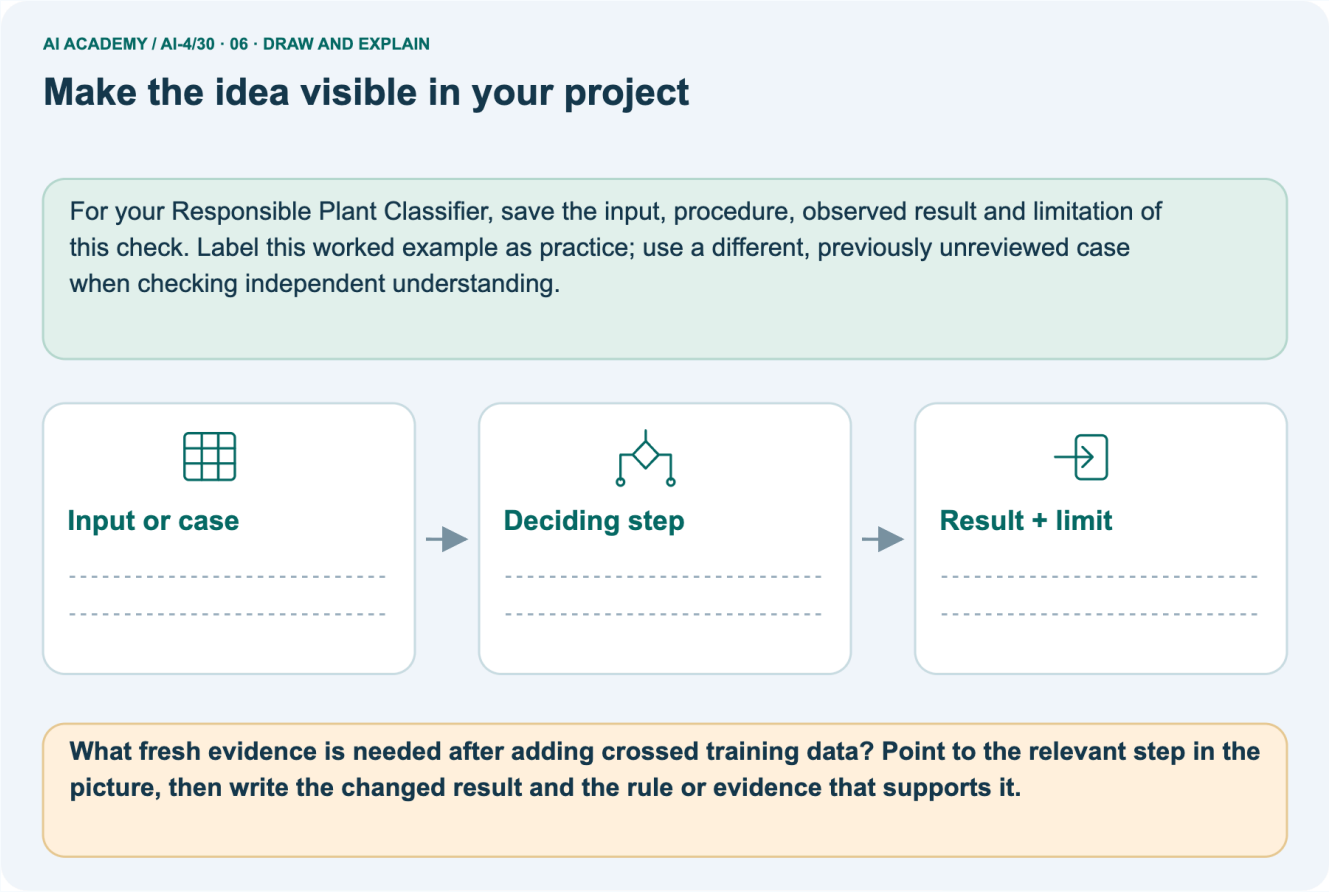
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Improve Generalization: the case

02. Follow the deciding steps

03. Overfitting and model capacity at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.